

MATH 6241 – Stochastic Processes

❖ Section 1. Stochastic Processes- Introduction

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- Stochastic Models
- Examples
- Monte Carlo Simulation
- Conditional Probability
- Conditional Expectation

□ Deterministic vs Stochastic systems

A **deterministic model** is $\{x_t\}_{t \in T}$ in which:

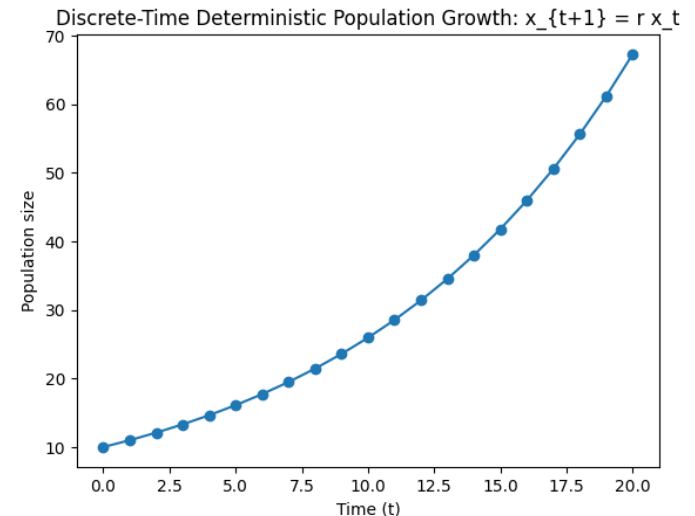
- There is **no randomness**
- The **future is completely determined** by the current state
- Given the same initial condition, the system always produces the **same trajectory**

Example: Discrete-Time Population Growth

Model: $x_{t+1} = r x_t$

Once you know initial population x_0 and growth rate r , then

$$x_t = r^t x_0 \text{ for all } t$$



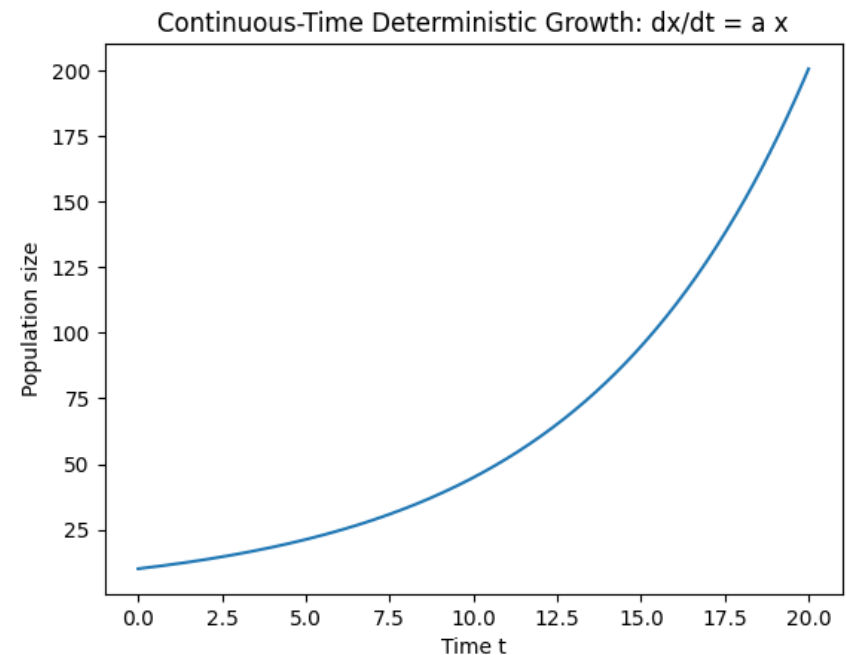
$$r = 1.1, \quad x_0 = 10$$

Example 2: Continuous-Time Growth

ODE Model : $x'(t) = ax(t)$

Given initial condition $x(0) = x_0$, and continuous growth rate a , then,

$$x(t) = x_0 e^{at}$$



Both of these models assume perfect predictability.
However, they are not realistic.

$$r = 0.15, \quad x_0 = 10$$

What is a stochastic process?

Instead of predicting *exactly what will happen*, stochastic process models describe what may happen and with what probability.

Definition: A **stochastic process** is a **collection of random variables** indexed by time:

$$\{X_t\}_{t \in T}$$

where,

- T is an index set.
- Each X_t is a **random variable** defined on the same probability space.

Discrete time: finite $T = \{0, 1, 2, \dots, N\}$ or infinite $T = \{0, 1, 2, \dots, t, \dots\}$

Continuous time: $T = [0, \infty)$

In discrete time case, we can write $\{X_t\}_{t \in T}$ as

$$\{X_0, X_1, X_2, \dots, X_N\} \quad \{X_0, X_1, X_2, \dots, X_t, \dots\}$$

The distribution of the random variable X_0 is called **initial distribution**.

Suppose sample space Ω is finite, the initial distribution with $P(X = j) = \alpha_j$ for $j = 1, 2, \dots, n$. We use the **row** vector notation:

$$\alpha = (\alpha_1, \dots, \alpha_n)$$

Review of terminology:

A Probability Space $(\Omega, \mathcal{F}, P)$ consists of three tightly connected objects:

1. **Sample Space Ω** , the set of **all possible outcomes** of the random experiment.
2. **Event Space $\mathcal{F}$ (σ -algebra)**, **collection of subsets of Ω** , called **events**.
3. **Probability Measure $P : \mathcal{F} \rightarrow [0,1]$**

A random variable is a function

$$X: \Omega \rightarrow \mathbb{R}$$

that assigns a numerical value to each outcome $\omega \in \Omega$.

Example: Binary i.i.d. Process

$$P(Y_t = 1) = p;$$

$$P(Y_t = -1) = 1 - p$$

$$p = 1/2$$

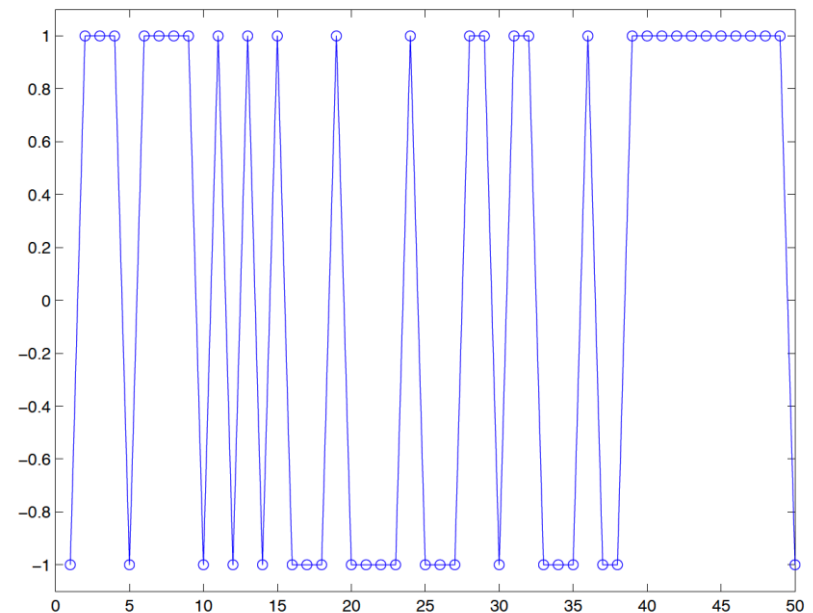
Flip a coin, you will get 1 *or* -1 dollar.

Now flip 50 times, we observed an outcome

$$\omega = (TFFFTTTFFFTT \dots FT)$$

$\Rightarrow$

$$Y_1(\omega) = -1 \quad Y_2(\omega) = 1$$

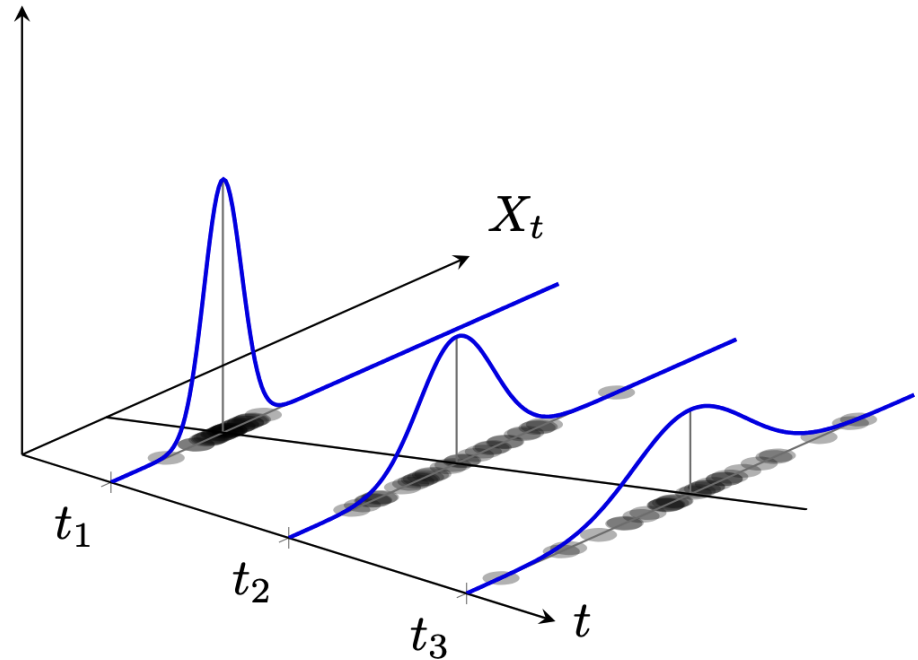


Three Ways to Think About a Stochastic Process

(1) For each fix time t ,

X_t is a random variable.

Mean and Variance of X_t are useful.



Example:

X_5 = number of customers at time 5

Y_{10} = stock price at time 10

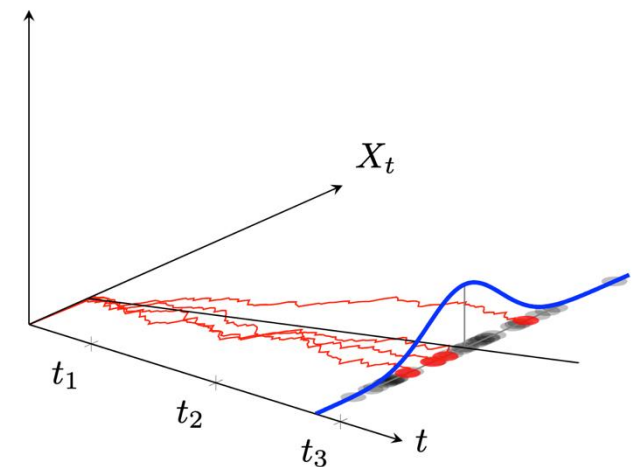
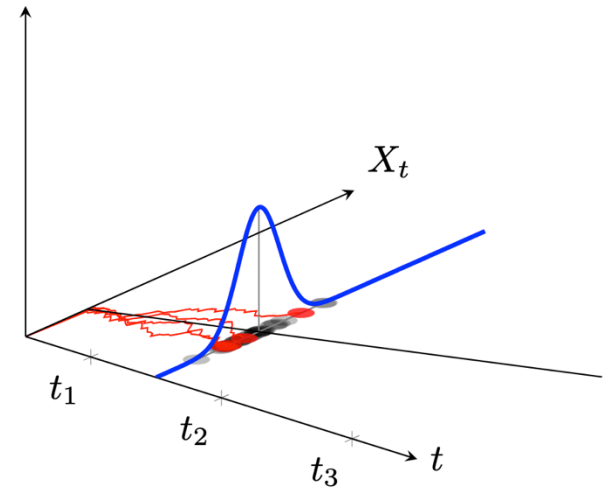
(2) For a fix outcome ω , it gives a trajectory or a sample path

$$t \rightarrow X_t(\omega)$$

Each curve is one realization $X_t(\omega)$:

$$X_0(\omega), X_1(\omega), X_2(\omega), \dots$$

which is a deterministic sequence.



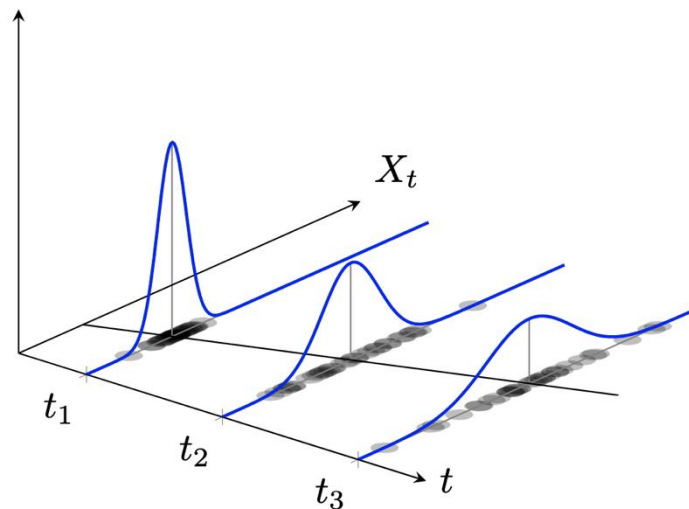
(3) For finitely many times $t_1, \dots, t_n$,

$$(X_{t_1}, \dots, X_{t_n})$$

has a joint probability distribution.

- IID case is too simple.
- Dependence between past and future make the model more realistic.

Example:



$$X_t \sim \text{Normal}(\mu_t, \sigma^2)$$

(X_{t_1}, X_{t_2}) is a multivariable normal distribution.

Deterministic process $x_0 \Rightarrow x_1 \Rightarrow x_2 \Rightarrow \dots$

Stochastic process: $X_0 \Rightarrow X_1 \Rightarrow X_2 \Rightarrow \dots$

Every stochastic process has:

- **Time index** (when things evolve)
- **State space** (possible values)

Time	State	Example
Discrete	Discrete	Markov chains
Discrete	Continuous	AR time series
Continuous	Discrete	Birth–death processes
Continuous	Continuous	Brownian motion

Example-Random Walks:

Let $X_0 = 0$ and define a random walk process:

$$X_t = \sum_{i=1}^t Y_i$$

Equivalently,

$$X_{t+1} = X_t + Y_{t+1}$$

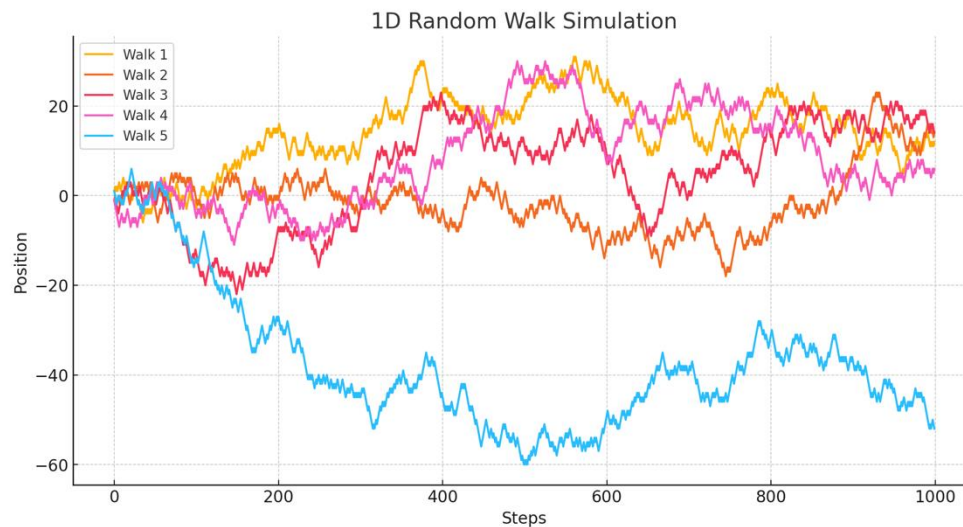
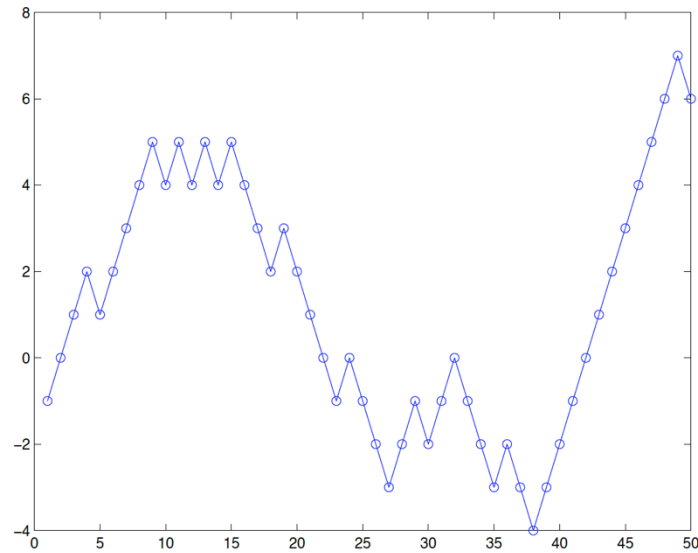
State Space

Mean $E(X_t)$

Variance $Var(X_t)$

Flip a coin t times, you will get 1 *or* -1 dollar each time.

$$\omega = (TFFFTTTFFFTT \dots FT) \Rightarrow X_{50}(\omega) = -1 + 1 + 1 + 1 - 1 + 1 + \dots + 1 - 1$$



Example: Gambler's ruins:

In each round, a gambler

- **win \$1** with probability p ,
- or **lose \$1** with probability $q = 1 - p$

The gambler starts with i dollars. The game stops when the gambler:

- Goes **broke** (capital 0), or
- Reaches a **target fortune** N



Stochastic Process (Random Walk) Model

X_t = gambler's capital after t rounds.

So, X_t can be modeled by random walk with bounds.

State space: $\{0, 1, 2, \dots, N\}$

Time: discrete $t = 0, 1, 2, \dots$

$$X_{t+1} = \begin{cases} X_t + 1 & \text{with probability } p \\ X_t - 1 & \text{with probability } q \end{cases}$$

Many problems in finance, biology, and queueing reduce to gambler's ruin.

Markov Property

The future depends only on the current capital:

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots) = P(X_{t+1} \mid X_t)$$

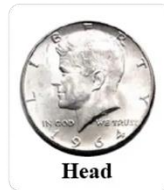
Example:

A coin toss game: two players flipping a fair coin,

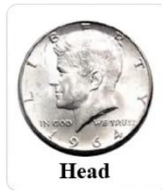
- Player A wins if the sequence **“HHT”** appears first.
- Player B wins if the sequence **“HTT”** appears first.

Is this a fair game?

A wins



Head



Head



Tail

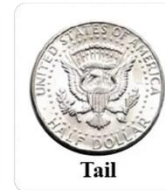
B wins



Head



Tail

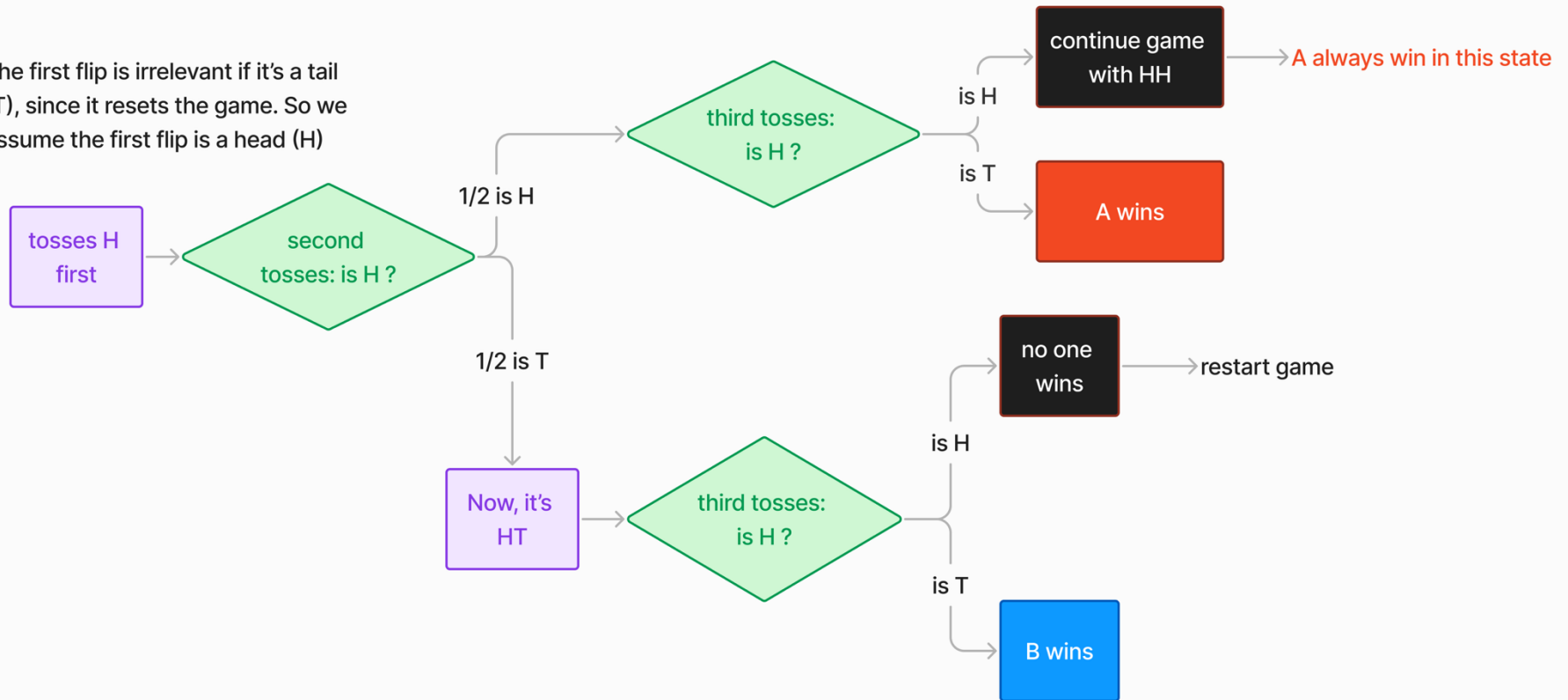


Tail

Method 1. Tree Diagram (Conditional Probability)

Flow chart

The first flip is irrelevant if it's a tail (T), since it resets the game. So we assume the first flip is a head (H)



We suppose the first toss is H .

Look at the second toss:

$$P = P(\text{Player A eventually win})$$

Case 1. Second toss is H

$$P(A \text{ wins} \mid HH) = 1$$

Case 2. Second toss is T

$$P(A \text{ wins} \mid HT) = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot P = \frac{1}{2}P$$

Combine the cases by **law of total probability**:

$$P = P(A \text{ wins} \mid HH)P(HH) + P(A \text{ wins} \mid HT)P(HT)$$

$$= 1 \times \frac{1}{2} + \frac{1}{2}P \times \frac{1}{2} = \frac{1}{2} + \frac{1}{4}P$$

$$\text{So, } P = \frac{2}{3}$$

Method 2, Markov Chain (later)

Model it as a Markov chain using the longest suffix of the current toss history that is also a prefix of one of the target patterns.

The game can be modeled as a Markov chain with states:

- S_0 : Start (no progress)
- S_1 : H (progress toward both patterns)
- S_2 : HH (progress toward HHT only)
- S_3 : HT (progress toward HTT only)

Absorbing outcomes:

- A wins if we reach **HHT**
- B wins if we reach **HTT**

Setting up the probability equations where $p_i = P(\text{A wins from state } i)$:
$$= P(\text{A wins} \mid \text{start in state } S_i).$$

Now write transition equations.

From S_0 :

- Toss H $\rightarrow S_1$
- Toss T $\rightarrow$ stay in S_0

$$p_0 = \frac{1}{2}p_1 + \frac{1}{2}p_0 \Rightarrow p_0 = p_1.$$

From S_1 (we have “H”):

- Toss H $\rightarrow S_2$
- Toss T $\rightarrow S_3$

$$p_1 = \frac{1}{2}p_2 + \frac{1}{2}p_3.$$

From S_2 (we have “HH”):

- Toss T $\rightarrow$ A wins immediately (HHT)
- Toss H $\rightarrow$ stays in S_2 (suffix is still “HH”)

$$p_2 = \frac{1}{2}(1) + \frac{1}{2}p_2.$$

Solve:

$$p_2 - \frac{1}{2}p_2 = \frac{1}{2} \Rightarrow \frac{1}{2}p_2 = \frac{1}{2} \Rightarrow p_2 = 1.$$

So once we reach HH , A is guaranteed to win.

From S_3 (we have “HT”):

- Toss T $\rightarrow$ B wins immediately (HTT), so A loses $\rightarrow$ probability 0
- Toss H $\rightarrow$ suffix becomes “H” $\rightarrow$ go to S_1

$$p_3 = \frac{1}{2}(0) + \frac{1}{2}p_1 = \frac{1}{2}p_1.$$

Now substitute everything into

$$p_1 = \frac{1}{2}p_2 + \frac{1}{2}p_3.$$

Using $p_2 = 1$ and $p_3 = \frac{1}{2}p_1$:

$$p_1 = \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{2}p_1\right) = \frac{1}{2} + \frac{1}{4}p_1.$$

Solve:

$$p_1 - \frac{1}{4}p_1 = \frac{1}{2} \Rightarrow \frac{3}{4}p_1 = \frac{1}{2} \Rightarrow p_1 = \frac{2}{3}.$$

Since $p_0 = p_1$,

$$p_0 = \frac{2}{3}.$$

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross